

12.421 / 12.621 Lecture 2

Maxwell's E_g and EM wave propagation

Topics covered

- Maxwell's E_g
- EM wave propagation in free space
- EM wave propagation in lossy materials
- Poynting vector in general media

- Review of constitutive parameters

Recall from last lecture that we can describe the EM properties of a material using 4 parameters, only 2 of which are relevant for RS

* $\epsilon' \epsilon_0$ electrical permittivity $\rightarrow$ store electric charge

* σ electrical conductivity $\rightarrow$ allow electric current

• μ magnetic permeability; $\mu = \mu_0 = \frac{\epsilon_0}{c^2} = 4\pi \times 10^{-7} \frac{H}{m}$
for virtually all RS applications

• ρ_v charge density; $\rho_v = 0$ in RS applications

We group the two important terms in the complex valued relative dielectric constant

$$\epsilon = \epsilon' - j\epsilon'' \quad \text{where} \quad \epsilon'' = \frac{\sigma}{\omega \epsilon_0}$$

- Maxwell's Eqs. - I assume that everyone has some familiarity with classical electrodynamics and the four fundamental equations often called Maxwell's Equations.

If these are totally new to you, consider consulting any number of excellent texts on the topic: Griffiths "Intro to electrodynamics", Purcell and Moren "Electricity and Magnetism", etc.

- Gauss's Law for electricity (where $\underline{E} \equiv$ electric field intensity vector, units: $\frac{V}{m}$)

$$\nabla \cdot \underline{E} = \frac{\rho_v}{\epsilon' \epsilon_0}$$

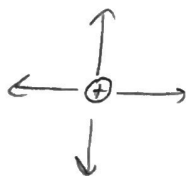
$$\text{where } \nabla = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right] = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

$$\Rightarrow \nabla \cdot \underline{E} \equiv \text{divergence of } \underline{E}$$

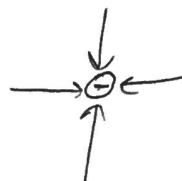
Review of divergence

- description of how much a vector field spreads (diverges) at a point

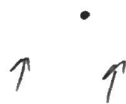
$$\nabla \cdot \underline{E} > 0$$



$$\nabla \cdot \underline{E} < 0$$



Zero divergence



- So, in essence, Gauss's Law says that a charge causes a flux through some closed surface
- In other words, like charges repel and the presence of a charge results in a diverging electric field
- As noted before, $\rho_v = 0$ in RS applications $\Rightarrow$ electric fields are divergence-free!
- An example of a divergence-free field:



12.421 / 12.621 Lecture 2

- Gauss's Law for magnetism

$$\nabla \cdot \underline{H} = 0 \quad H \equiv \text{magnetic field intensity vector, units } \frac{A}{m}$$

• Fun fact, there is no known magnetic "charge"

$\Rightarrow$ magnetic fields are always divergence free and tend to form closed loops

- Faraday's Law

$$\nabla \times \underline{E} = -\mu \frac{\partial \underline{H}}{\partial t} \quad \text{where } \nabla \times \underline{E} \equiv \text{curl of } \underline{E}$$

Review of curl

$$\nabla \times \underline{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

$$= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{x} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{y} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{z}$$

Measure of rotation (curl) of vector field about a point

- Time-varying magnetic field induces an electric field, and vice-versa
- Also gives confirmation from Gauss's Law of electricity that in RS applications ($\rho_v = 0$), E field forms closed loops (zero divergence, non-zero curl)

- Ampere's Law (w/ Maxwell's correction)

$$\nabla \times \underline{H} = \epsilon' \epsilon_0 \frac{\partial \underline{E}}{\partial t} + \sigma \underline{E}$$

↑ This was Maxwell's correction, and it leads directly to one of the greatest discoveries ... as we'll soon see

• Tells us there are 2 ways to generate a magnetic field

1. time-varying $\underline{E}$ field

2. electric current

• For $\frac{\partial \underline{E}}{\partial t} = 0$



But when we have $\frac{\partial \underline{E}}{\partial t} \neq 0$, Faraday's Law and Ampere's Law work together (rearranging for illustrative purposes)

$$\nabla \times \underline{E} = -\mu \frac{\partial \underline{H}}{\partial t}$$

$$\nabla \times \underline{H} = \epsilon' \epsilon_0 \frac{\partial \underline{E}}{\partial t} + \sigma \underline{E}$$

oscillating $\underline{E}$ and $\underline{H}$ fields

$\Rightarrow$ propagation of a wave $\rightarrow$ will go on forever unless $\sigma > 0$

12.421/12.621 Lecture 2

• EM wave eq. in free space

- In free space, $\sigma=0$, $\mu=\mu_0$, $\epsilon'=1$, $\rho_v=0$

$$\nabla \times \underline{E} = -\mu_0 \frac{\partial \underline{H}}{\partial t}$$

$$\nabla \times \underline{H} = \epsilon_0 \frac{\partial \underline{E}}{\partial t}$$

For E field, take curl of both sides of Faraday's Law

$$\nabla \times \nabla \times \underline{E} = -\mu_0 \nabla \times \left(\frac{\partial \underline{H}}{\partial t} \right)$$

apply the identity

$$\nabla \times \nabla \times \underline{E} = \nabla (\nabla \cdot \underline{E}) - \nabla^2 \underline{E} \quad (\nabla^2 = \nabla \cdot \nabla \equiv \text{Laplacian})$$

Recall that Gauss's Law is $\nabla \cdot \underline{E} = 0$

$$\Rightarrow \nabla^2 \underline{E} = \mu_0 \frac{\partial}{\partial t} (\nabla \times \underline{H})$$

Plugging in Ampere's Law

$$\nabla^2 \underline{E} = \mu_0 \epsilon_0 \frac{\partial^2 \underline{E}}{\partial t^2} \quad \rightarrow \text{EM wave equation in free space}$$

What does this tell us?

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \equiv \text{wavespeed} \rightarrow \text{speed of light}$$

$\hookrightarrow$ Because μ_0 and ϵ_0 are universal constants, c is constant for all EM waves in free space

• EM wave free space (cont.)

- H field : we can take the same approach to show

$$\nabla^2 \underline{H} = \mu_0 \epsilon_0 \frac{\partial^2 \underline{H}}{\partial t^2}$$

- Now, let's correct these ideas.

- Consider a uniform plane wave propagating along $+\hat{z}$

- For convenience, let us define the electric field in terms of a phasor (phase vector) that depends on spatial coordinates x, y, z (real component of complex value)

$$\underline{E}(x, y, z, t) = \text{Re} \left[\underline{E}(x, y, z) e^{j\omega t} \right]$$

and similarly for $\underline{H}$ (recall, $\omega \equiv$ angular freq)

- Thus, we have

$$\underline{E} = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$$

- And the wave eq gives

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + K_0^2 \right) E_x = 0$$

where $K_0 = \frac{\omega}{c} = \frac{2\pi}{\lambda_0} \equiv$ angular wavenumber in free space ($\lambda_0 \equiv$ wave length in free sp)

- We henceforth define our coordinate system such that

$$E_y = 0$$

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- Because we are considering a ^{uniform} plane wave propagating along $\hat{z}$, we have $\frac{\partial^2 E_x}{\partial x^2} = \frac{\partial^2 E_x}{\partial y^2} = 0$
 $\Rightarrow$ wave eq gives

$$\frac{\partial^2 E_x}{\partial z^2} + k_0^2 E_x = 0$$

for which, the general solution is

$$E_x(z) = E_{x_0} e^{-jk_0 z}$$

or

$$\underline{E}(z) = E_{x_0} e^{-jk_0 z} \hat{x}$$

- We can then solve for corresponding magnetic field by plugging into Faraday's Law

$$\underline{H}(z) = \frac{1}{\eta_0} E_{x_0} e^{-jk_0 z} \hat{y}$$

where $\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \equiv$ intrinsic impedance of free space
 $\approx 377 \Omega$

- For the purposes of comparison, let's rewrite these expressions for $\underline{E}$ and $\underline{H}$

$$\begin{aligned} \underline{E}(z) &= E_{x_0} e^{-jk_0 z} \hat{x} \\ \underline{H}(z) &= \frac{1}{\eta_0} E_{x_0} e^{-jk_0 z} \hat{y} \end{aligned}$$

• EM wave in free sp (cont.)

- What do these eggs tell us?

1. Remind us that $\underline{E}$ and $\underline{H}$ fields are a package deal, can't have one without the other (although, this was clear from Maxwell's Eggs)
2. Show us that $\underline{H} \perp \underline{E}$ (at least in free space, though this is generalizable to any medium as we shall see $\Rightarrow$ $\underline{H}$ and $\underline{E}$ are always orthogonal)
3. $\underline{E}$ and $\underline{H}$ are in phase, but this is not a general property; as we will show, conductivity causes a phase shift
4. $\underline{E}$ and $\underline{H}$ are related through intrinsic impedance

- It can be shown that in 3D

$$\underline{E} = -\eta (\hat{k} \times \underline{H})$$

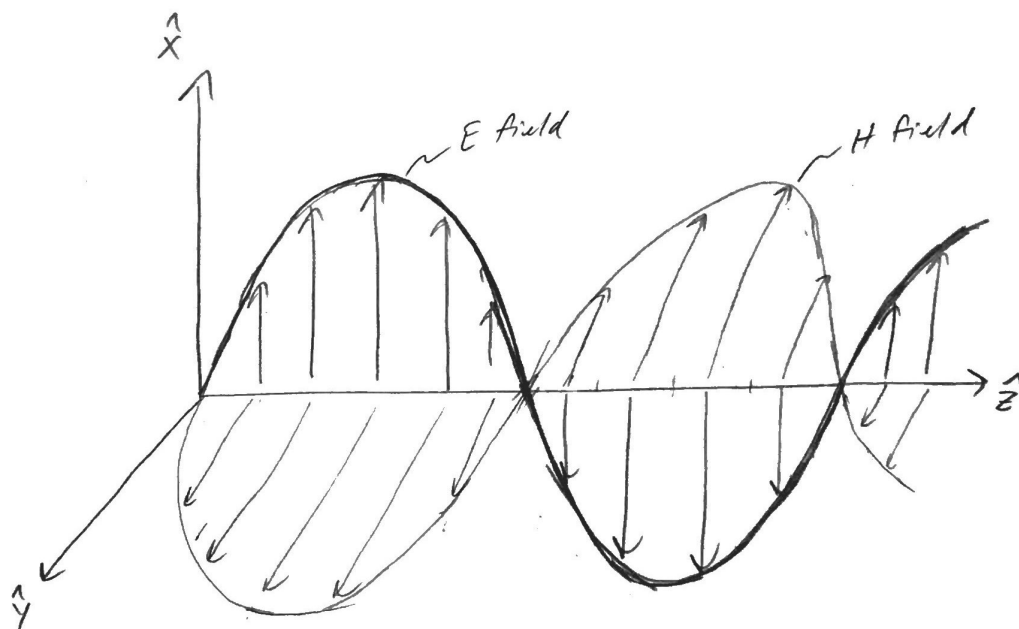
$$\underline{H} = \frac{1}{\eta} (\hat{k} \times \underline{E})$$

where $\hat{k} \equiv$ propagation unit vector

$\uparrow$ This is true for any material $\checkmark$
appropriate definition of η

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- The preceding equations make clear that EM waves are "transverse waves" meaning they oscillate only perpendicular to the direction of propagation $\rightarrow$ a key distinction between EM waves and mechanical waves (e.g., sound waves), which can have longitudinal components



• EM waves in homogeneous, isotropic, non-ferromagnetic material

- Now we have $\sigma \geq 0$, $\epsilon' \geq 1$, $\mu = \mu_0$, $\rho_v = 0$

- In free space, we saw that all EM waves travel at the same speed and will travel forever (no attenuation)

- What about other materials?

- Recall our form for E field intensity

$$\underline{E}(x, y, z, t) = \text{Re}[\underline{E}(x, y, z) e^{j\omega t}]$$

- Ampere's Law

$$\begin{aligned}\nabla \times \underline{H} &= \epsilon' \epsilon_0 \frac{\partial \underline{E}}{\partial t} + \sigma \underline{E} \\ &= j\omega \epsilon' \epsilon_0 \underline{E} + \sigma \underline{E} \\ &= j\omega \epsilon_0 \underbrace{\left(\epsilon' - j \frac{\sigma}{\omega \epsilon_0} \right)}_{\text{dielectric constant } \epsilon} \underline{E} \\ &= j\omega \epsilon \epsilon_0 \underline{E}\end{aligned}$$

• Faraday's Law

$$\nabla \times \underline{E} = -j\omega \mu_0 \underline{H}$$

• Taking the curl at both sides of Faraday's Law gives

$$\nabla^2 \underline{E} - \gamma^2 \underline{E} = 0$$

wave eq in general media

$$\gamma = j\omega \sqrt{\mu_0 \epsilon \epsilon_0}$$

complex-valued propagation constant

↳ can derive similar expression for H field

- Case of low-loss or lossless media ($\epsilon \approx \epsilon'$)

• Define wave number as

$$K \equiv \frac{2\pi}{\lambda} = -j\gamma = \omega \sqrt{\mu_0 \epsilon' \epsilon_0} = \frac{\omega}{c} \sqrt{\epsilon'} = K_0 \sqrt{\epsilon'}$$

• Thus, the phase velocity of the EM wave is now

$$v_p = \frac{\omega}{K} = \frac{c}{\sqrt{\epsilon'}} \quad \left(\text{also common to use } n' = \sqrt{\epsilon'} = \frac{c}{v_p} \right)$$

where $n \equiv \text{index of refraction}$

↳ phase velocity shows a couple of important points

$$1. v_p = v_p(\omega) \quad \text{when } \epsilon' = \epsilon(\omega)$$

=> common for EM waves to be dispersive (waves of different frequencies propagate at different speeds)

$$2. v_p < c \quad \text{for } \epsilon' > 1$$

• Now, following similar steps as before, guess the wave eq. as

$$\frac{\partial^2 E_x}{\partial z^2} + K^2 E_x = 0$$

solution takes the form

$$\underline{E}(z) = E_{x0} e^{-jKz} \hat{x}$$

$$\underline{H}(z) = \frac{1}{\eta} E_x e^{-jKz} \hat{y}$$

$$\text{where } \eta = \frac{\omega \mu_0}{K} = \sqrt{\frac{\mu_0}{\epsilon' \epsilon_0}} = \frac{\mu_0}{\sqrt{\epsilon'}} \hat{y}$$

* Because η is real, E and H are in phase

12.421 / 12.621 Lecture 2

Let's add conduction to the mix!

- Case of lossy materials

• Recall wave eq.

$$\nabla^2 \underline{E} + \gamma^2 \underline{E} = 0 \quad \text{and} \quad \nabla^2 \underline{H} + \gamma^2 \underline{H} = 0$$

$$\text{where } \gamma^2 = -\omega^2 \mu_0 \epsilon \epsilon_0 = -k_0^2 \epsilon$$

• For convenience (and convention), let's define

$$\gamma = \alpha + i\beta$$

$\alpha \equiv$ attenuation const

$\beta \equiv$ phase const

• Uniform plane wave propagating along $\hat{z}$ with E field oriented along $\hat{x}$

$$\underline{E}(z) = \hat{x} E_{x_0} e^{-\gamma z} = \hat{x} E_{x_0} e^{-\alpha z} e^{-j\beta z}$$

β acts as wave number

α attenuates = intensity decays by $\frac{1}{e}$ over distance $\delta_s = \frac{1}{\alpha}$

$$\underline{H}(z) = \hat{y} \frac{E_{x_0}}{\eta} e^{-\alpha z} e^{-j\beta z}$$

$\delta_s \equiv$ skin depth

$= 2\delta_p$ $\delta_p \equiv$ penetration depth

$$\text{where } \eta = \frac{\eta_0}{\sqrt{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'}\right)^{-1/2}$$

This shows us that when $\epsilon'' > 0$ (ie., $\sigma > 0$),

η is complex-valued $\Rightarrow$ phase shift between E and H fields

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- Expanding α and β gives

$$\alpha = -\omega \sqrt{\mu_0 \epsilon_0} \operatorname{Im}(\sqrt{\epsilon^r})$$

$$= K_0 \left\{ \frac{\epsilon'}{2} \left[-1 + \sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} \right] \right\}^{1/2}$$

$$\beta = \omega \sqrt{\mu_0 \epsilon_0} \operatorname{Re}(\sqrt{\epsilon^r})$$

$$= K_0 \left\{ \frac{\epsilon'}{2} \left[1 + \sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} \right] \right\}^{1/2}$$

sanity check:
 $\alpha = 0$ when $\epsilon'' = 0$

↑
 compare w/ lossless
 case from before

↓
 sanity check
 $\beta = K_0 \sqrt{\epsilon'}$ when
 $\epsilon'' = 0$

- Connecting to index of refraction (commonly used in optics and many other fields)

$$n = n' - j n''$$

$$= \operatorname{Re}(\sqrt{\epsilon^r}) - j \operatorname{Im}(\sqrt{\epsilon^r})$$

$$\Rightarrow \alpha = K_0 n''$$

$$\beta = K_0 n'$$

- EM Power density - Poynting vector $\underline{S}$

- Flow of power (i.e., energy flux) carried by EM wave
 given by Poynting vector

$$\underline{S}(t) = \underline{E}(t) \times \underline{H}(t) \quad \left[\frac{\text{W}}{\text{m}^2} \right]$$

- Averaged over time

$$\underline{S} = \frac{1}{2} \operatorname{Re}[\underline{E} \times \underline{H}^*] = \frac{1}{2} \frac{|E_0|^2}{2|\eta|} e^{-2\alpha z} \cos \theta_\eta$$

note factor of 2 for power

→ phase of η

uniform plane wave propagating along $\hat{z}$